



Risks and burden of lung cancer incidence for residential petrochemical industrial complexes: A meta-analysis and application

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ABSTRACT

Background: Lung cancer is one of the most common cancers in the world. Higher incidence of lung cancer may be associated with residential proximity to a petrochemical industrial complex (PIC) due to exposure to various carcinogens, although results from previous epidemiologic studies remain inconclusive. Because disease burden due to residential inequality is a public health and societal concern, this study analyzed published data to estimate lung cancer incidence in association with residential proximity to PICs.

Methods: We performed a meta-analysis on selected epidemiologic studies that met the following criteria: lung cancer incidence was coded by the International Classification of Diseases; exposure groups were clearly defined as residents living near PICs; and confidence intervals were available or calculable from original articles. We further applied a population attributable factor (PAF) method to estimate disease burden attributable to living near PICs in 22 European Union (EU) countries.

Results: Meta-analysis included six studies with a total of 466,066 residents living near PICs in six countries. Residents living near PICs had a 19% higher risk of lung cancer compared to those who lived farther away (95% CI = 1.06–1.32). By sex, risks were higher and more significant for females (RR = 1.29; 95% CI = 1.09–1.54; $P = 0.004$) than males (RR = 1.12; 95% CI = 0.95–1.33; $P = 0.173$). By location, only groups in Europe had a significantly greater risk of lung cancer with exposure to PICs (95% CI = 1.03–1.33; $P = 0.019$), although groups in other locations showed similar trends. By bona fide observation, observation of residents for at least seven years provided sufficient latency to estimate risk (RR = 1.25; 95% CI = 1.17–1.34; $P < 0.001$). Regarding burden of lung cancer in 22 EU countries, 494 males and 478 females were attributed to living in the vicinity of a PIC annually.

Conclusions: Lung cancer incidence is significantly higher in individuals living near PICs. This result provides strong epidemiologic evidence for further policy to regulate potential pollutants near PICs.

Highlights: Higher incident rates of lung cancer for residents living close to petrochemical industry complex

1. Introduction

Industrial development, which generally refers to chemical production and consumption, is associated with the emergence of environmental risk factors due to industrial release of hazardous chemicals into the environment (Elliott et al., 2004; Zhang et al., 2010). Estimating health risks associated with environmental exposure is

important for residential justice and to identify strategies to reduce disease burden and health inequality (Elliott et al., 2004; Solomon et al., 2016).

Since the mid-1970s, there has been growing concern over excessive cancer cases among residents living in coastal counties in the United States (U.S.) (Blot and Fraumeni, 1976). One major concern is the impact of rapid development of petrochemical industrial complexes

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(PICs)—defined as clusters of industries that manufacture petroleum and chemical products (United Nations Statistics Division (UNSD), 2008)—on the long-term health of residents in the vicinity of these areas (Lin et al., 2017; Lopez-Navarro et al., 2013; Nadal et al., 2011). Industrial PIC manufacturing processes are diverse and usually involve toxic chemicals, such as volatile organic compounds, polycyclic aromatic hydrocarbons, polychlorinated biphenyls, ethylene, propane, polyvinyl chloride, benzene, and heavy metals (Axelsson et al., 2010; Beccaloni et al., 2014; Li et al., 2009; Cetin et al., 2003). Exposure to these toxicants among residents living close to PICs can increase their risk of cancers, particularly respiratory cancers such as lung cancer (Kaldor et al., 1984).

As one of the most common cancers worldwide, lung cancer has global incidence rates of 36.75 and 17.41 per 100,000 males and females, respectively, contributing to around 682,000 cases globally in 2016 (Global Burden of Disease Collaborative Network, 2016). Apart from well-known behavioral risk factors, such as smoking, and naturally occurring environmental carcinogens, such as radon, other environmental determinants also contribute to lung cancer (Li et al., 2009; International Energy Agency, 2017; Tirmarche et al., 2010).

Due to public health concerns, several studies have attempted to address whether there is an increased risk of lung cancer in residents living near PICs, but results have varied across countries and study periods. In the 1980s, Kaldor and colleagues found a higher risk of lung cancer among males living close to PICs (Kaldor et al., 1984). Some U.S. studies also have reported consistent findings of an association between lung cancer risk and residential exposure to PICs (Simonsen et al., 2010; Gottlieb et al., 1982). However, Anna-Karin and colleagues' study in Ecuador reported statistical non-significance of the relative risks of lung cancer with residential exposure to PICs (Hurtig and San Sebastian, 2002). One Italian study even reported a protective effect of residential PIC exposure for females in Melilli and Priolo, although the results were not statistically significant (Fazzo et al., 2016).

Our previous research reported a higher risk of lung cancer mortality among residents living near PICs, although these results also were not statistically significant (Lin et al., 2017). To our best knowledge, it is the only study to focus on lung cancer mortality. Further, no studies have systematically combined previous data to quantitatively examine the risk of lung cancer incidence among residents living near PICs. Therefore, using a random-effects model in meta-analysis, we aimed to examine heterogeneity among previous studies and estimate the pooled risk of lung cancer incidence among residents living near PICs, including both environmental exposure and co-exposure from occupation. Since most studied groups were from European countries, and PIC data from the European Union (EU) are most accessible, we further estimated the burden of lung cancer attributable to PICs in 22 EU countries.

2. Methods

2.1. Literature review and study assessment

Two authors (RT Lin and YT Hsu) independently searched published studies available before October 16, 2017, from PubMed, Web of Science, Cochrane Library, ScienceDirect, and other sources. Search terms were: “(lung cancer OR lung neoplasm) AND (refinery OR petroleum OR petrochemical OR oil and gas industry).” The authors screened articles and included studies that met the following criteria: (1) original article available in English and full-text; and (2) title and abstract related to the study of interest.

We included studies in our analysis if they: (1) reported data on lung cancer incidence (as the outcome variable); (2) defined exposure group as residents living in the vicinity of PICs (with the definition of PIC exposure); and (3) did not contain populations overlapping with other studies. Studies were excluded if they focused only on occupational exposure or if point estimates and/or confidence intervals (CI) were not

extractable or calculable.

We then applied the Newcastle-Ottawa Quality Assessment Scale (NOQAS) to assess study quality and possible risk of bias. NOQAS rates eight items in three categories—selection, comparability, and outcome—with nine stars to semi-quantitatively evaluate study quality (Wells et al., n.d.). We reviewed study quality independently and assigned stars (N-O score) for each item.

2.2. Data combination and synthesis

Four types of point estimates were reported in the selected studies: odds ratio (OR), relative risk (RR), standardized incidence ratio (SIR), and age-standardized rate ratio (ASRR). SIR or ASRR represent a comparison of the observed incidence in a studied group to the expected incidence in an age- and sex-adjusted standard population (Axelsson et al., 2010; Fazzo et al., 2016; Zusman et al., 2012). OR, ASRR, and SIR can be interpreted and analyzed as approximates of RR (Symons and Taulbee, 1981).

For studies that did not report either 95% CI or standard error (SE) (Kaldor et al., 1984; Zusman et al., 2012; Fares and Masri, 2012), we estimated variances and SE of $\ln RR$ using the following equation:

$$\begin{aligned} Var(\ln RR) &= Var(\ln R_1) + Var(\ln R_0) \\ &\approx \left(\frac{1}{R_1}\right)^2 \times \frac{R_1(1 - R_1)}{N_1} + \left(\frac{1}{R_0}\right)^2 \times \frac{R_0(1 - R_0)}{N_0} \end{aligned} \quad (1)$$

where $Var(\ln RR)$ represents the variance of natural log of RR; R_1 and R_0 represent incidence rates of the studied group and reference group, respectively; and N_1 and N_0 represent populations of the studied group and reference group, respectively.

For studies that only provided upper and lower limits of 95% CI (Axelsson et al., 2010; Simonsen et al., 2010; Hurtig and San Sebastian, 2002; Fazzo et al., 2016), we estimated SE of natural log of RR using the following equation:

$$SE(\ln RR) = \frac{1}{2} \times \left(\frac{\ln(RR_{upper}) - \ln(RR_{est})}{1.96} + \frac{\ln(RR_{est}) - \ln(RR_{lower})}{1.96} \right) \quad (2)$$

where $\ln(RR_{upper})$, $\ln(RR_{est})$, and $\ln(RR_{lower})$ represent the natural logs of upper limit, point estimate, and lower limit of RR, respectively.

2.3. Statistical analysis

Pooled point estimates with 95% CI were analyzed by a random effect model. We applied the I^2 test to address possible heterogeneity within and between studies by defining 10% as no heterogeneity, 10%–30% as low heterogeneity, 30%–60% as moderate heterogeneity, and > 60% as high heterogeneity (Zhao, 2013). Subgroup analysis was also performed to examine the effects among groups stratified by characteristics, including sex, location, and bona fide observation, which was defined as a study period of < 7 or ≥ 7 years after 20 years of PIC operation. Exposures to volatile organic compounds (VOC), polycyclic aromatic hydrocarbon (PAH), benzene, benzo-a-pyrene (BAP), heavy metal, and polychlorinated biphenyls (PCB) well documented from original researches were extracted for subgroup analysis.

We applied meta-regressions to investigate possible factors of heterogeneity, including latency of study period, starting year of study, ending year of study, published year, and starting year of PIC operation. By continually adding one estimate to the pooled estimate in order of starting year of study, we provided another approach for sensitivity analysis. Finally, we used a funnel plot to examine whether there was publication bias and used Begg's and Egger's regressions to test whether there was bias due to a small-study effect.

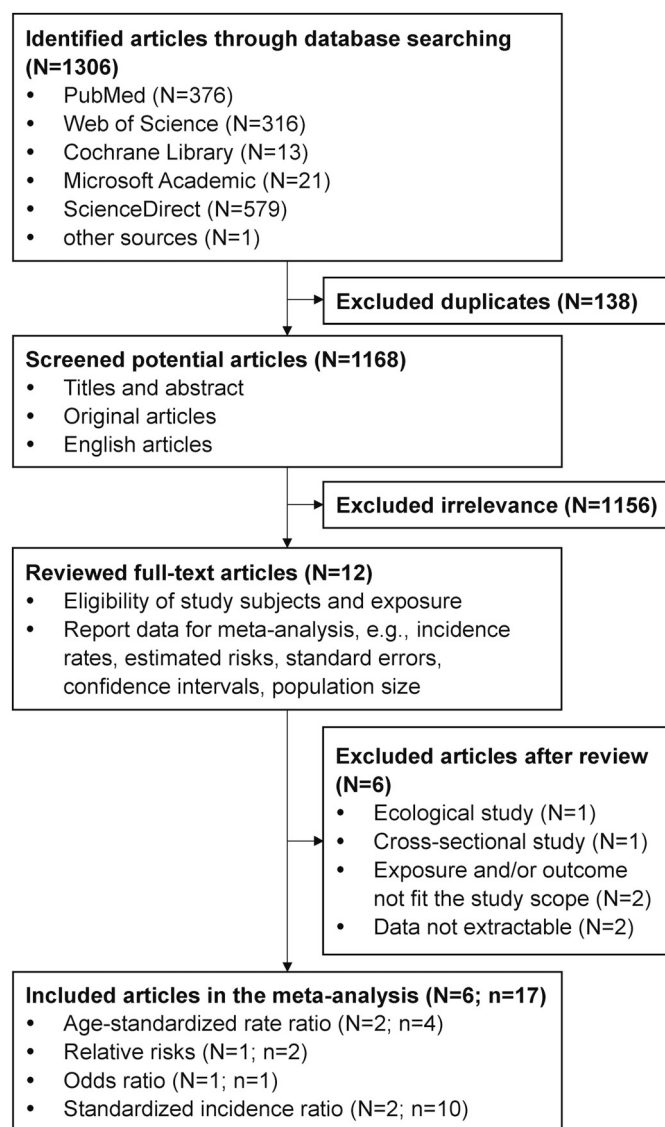


Fig. 1. Study identification, screening, review, and selection. N = number of studies; n = number of estimates included into meta-analysis.

2.4. Estimating burden of lung cancer incidence

To estimate the cases of lung cancer attributable to PICs in 22 EU countries, we calculated population attributable factor (PAF) by applying the estimated RR. The formula for PAF is written as follows (Levin, 1953):

$$PAF_{ij} = \frac{P_i \times (RR_j - 1)}{1 + P_i \times (RR_j - 1)} \quad (3)$$

where P_i is proportion of the exposed population living in the vicinity of PICs for a given country i , obtained from literature report (Lukach et al., 2015), assuming equal sex distribution, and RR_j is relative risk for sex j (males or females), obtained from the pooled estimate. Disease burden was calculated by multiplying lung cancer cases by the corresponding PAF, where lung cancer incidence and 95% CI were estimated from the Global Burden Disease Study (Global Burden of Disease Collaborative Network, 2016).

All analyses were performed using Stata Software version 11.2 (StataCorp, TX, US). $P < 0.05$ was considered statistically significant for two-sided tests.

3. Results

Fig. 1 shows the process of study identification, screening, review, and selection. A complete search identified 1306 articles from six databases in the initial stage. Full texts of twelve studies were read carefully and then 6 were excluded after further review, leaving 6 studies for analysis. Therefore, the pooled analysis included six articles covering 17 studied groups, with four reporting point estimates in ASRR, two in RRs, one in ORs, and ten in SIRs.

Table 1 lists the basic characteristics of groups analyzed in the published studies included in our meta-analysis. The 17 groups covered ~466,066 individuals living near PICs in six countries: U.S., Ecuador, Sweden, Israel, Italy, and Serbia. Except for one case-control study (study ID = A in Table 1), all studies were cohort studies. All reported clear definition and measurement of PIC exposure. The study period encompassed 38 years (1974–2006). The shortest interval between the start of PIC operation and starting year of study was 11 years, rendering enough latency for every study.

Fig. 2 shows pooled estimates of the risk for lung cancer incidence among residents living close to PICs for each study group. An overall pooled RR of 1.19 (95% CI = 1.06–1.32) indicated that lung cancer incidence was significantly associated with exposure to PICs, despite five out of 17 studied groups reporting point estimates of < 1.0 . Regarding the influence of each studied group on the overall meta-analysis, the study of males in Augusta, Italy (study ID = N) accounted for the largest proportion (22.63%), mainly due to its narrowest 95% CI (1.13–1.37). Although the study in Ecuador (study ID = C) reported the highest point estimate of 1.65, it contributed to only 0.30% of the overall meta-analysis. The overall I^2 was 28.6%, indicating low heterogeneity among studied groups.

To further characterize risks across diverse groups, we conducted subgroup analyses stratified by sex, location, and bona fide observation (Table 2). When stratified by sex, only females had a significantly greater risk of lung cancer with exposure to PICs (pooled RR = 1.29; 95% CI = 1.09–1.54; $P = 0.004$), although a similar trend was found for males (pooled RR = 1.12; 95% CI = 0.95–1.33; $P = 0.173$). When stratified by location, only groups in Europe had a significant, 1.17-fold greater risk of lung cancer with exposure to PICs (95% CI = 1.03–1.33; $P = 0.019$), although groups in other locations had positive risks that were not statistically significant. When stratified by bona fide observation, groups with longer observation periods (≥ 7 years) had significantly higher risk (pooled RR = 1.25; 95% CI = 1.17–1.34; $P < 0.001$), indicating that a longer latency period might provide more sufficient time to assess lung cancer incidence. Significant higher risks of lung cancer are also found among PIC processing chemicals of VOC, PAH, BAP, heavy metal, and PCB (RR = 1.24). Residents living close to PIC with possible exposure to benzene also have higher, but not significant risk of lung cancer (RR = 1.06, 95% CI = 0.74–1.53).

In sensitivity analysis, we gradually added groups to the meta-analysis by starting with the earliest starting year of study. Groups with the same starting year were added randomly. No studied groups significantly affected the results of pooled estimates (Fig. 3). We did not find any significant factors of heterogeneity in meta-regressions. Among the 17 studied groups, no significant publication bias was found (Egger's test, $P = 0.183$; Begg's test, $P = 0.108$). Symmetric distribution of a funnel plot of the 17 studied groups also demonstrated that publication bias was unlikely (Supplementary Fig. S1).

Supplementary Section 3 and Section 4 list details of quality assessment for the five cohort studies and one case-control study, respectively. Zusman's study received maximal N-O scores (Zusman et al., 2012), as did the only case-control study. The lowest N-O score of all cohort studies was seven out of nine (Axelsson et al., 2010; Hurtig and San Sebastian, 2002; Bulat et al., 2011), and its low score was due to losing stars on the items “selection of the non-exposed cohort” in the selection category and “adequacy of follow up of cohort” in the outcome category (Supplementary Table S1). However, those studies did

Table 1
Basic characteristics of studies included in the meta-analysis.

ID	Comparison	Study period	The started operation year of petrochemical plant	Adjusted confounders	Country	Definition of petroleum area and reference area	Study group (number of subject)	Outcome selection	Industrial activity/substantial chemicals	Study design	Reference
A	Both sexes, ever living within 0.5 mile buffer vs. never living within 0.5 mile buffer	1998–2001	1901	Age, sex, race, education, metropolitan residence, and smoking history	US (Louisiana)	Petroleum area: 11 parishes classified as petroleum sites in Louisiana based on the Standard Industrial Classification codes (2865, 2869, 5169, 2812, 2899, 2911, 2821, 2851, 2822), including East Baton Rouge, West Baton Rouge, Iberville, Ascension, St. John, St. James, St. Charles, Orleans, Jefferson, St. Bernard and Plaquemine parishes (N = 437) Petroleum area: Four counties (Lago Agrio, Shushufindi, Orellana, and Sachas) where oil exploitation had been ongoing for > 20 years before study started.	Exposure group: Residents aged 20–74 in one of the 11 parishes at the time of diagnosis of lung cancer (N = 455) Reference group: Random sample of residents matched to lung cancer cases on age, sex, and race who didn't have lung cancer in the 11 parishes (N = 437)	Reported indicator: Adjusted odds ratio ICD version and code: ICD-9, 162.2–162.9 Source: Louisiana Tumor Registry	Hydrocarbons (HC)	Case-control study	Simonsen et al., 2010
B	Male indigenous people and peasants, exposed counties vs. non-exposed counties	1985–1998	1972	Age, sex	Ecuador (Sucumbios, Orellana, Napo, and Pastaza)	Counties (Lago Agrio, Shushufindi, Orellana, and Sachas) where oil exploitation had been ongoing for > 20 years before study started.	Exposure group: Male residents of the petroleum areas (N = 65,045) Reference group: Male residents of the reference areas (N = 81,592)	Reported indicator: Relative risks, derived from the ratio of age-adjusted incidence rates in the exposure group vs. reference group	HC, volatile organic compounds (VOCs, e.g., benzene, xylene, and toluene), and polycyclic aromatic hydrocarbons (PAH)	Cohort study	Hurtig and San Sebastian, 2002
C	Female indigenous people and peasants, exposed counties vs. non-exposed counties	1985–1998	1972	Age, sex	Ecuador (Sucumbios, Orellana, Napo, and Pastaza)	Counties without oil developing activities (Cascales, Pro El Carmen, La Bonita, Lumbabiqui, Aquarico, Tena, Archidona, El Chaco, Baeza, Puyo, and Mera).	Exposure group: Female residents of the petroleum areas (N = 53,219) Reference group: Female residents of the reference areas (N = 74,118)	ICD version and code: ICD-10, C34 Source: National Cancer Registry (Cases from the Sucumbios, Orellana, Napo, and Pastaza were diagnosed in Quito)		Cohort study	Hurtig and San Sebastian, 2002
D	Males, area with petrochemical industry vs. areas without petrochemical industry	1974–2005	1963	Age, sex	Sweden (Norum, located in central Stenungsund)	Petroleum area: Norum municipalities without petrochemical industry located in the western Sweden Reference area: Five municipalities without petrochemical industry located in the western Sweden	Exposure group: Male residents in area with petrochemical industry (N = N.R.) Reference group: Male residents in area without petrochemical industry (N = N.R.)	Reported indicator: Standardized incidence ratio ICD version and code: ICD-7, 162.0–162.1 Source: Regional cancer registry (Norum, Stenungsund)	Ethylene, propene, polyethylene, poly vinyl chloride (PVC), amines, surfactants, and oxo-alcohols, benzene, 1,3-butadiene, ethylene oxide, vinyl chloride (VCM), 1,2-dichloroethane (EDC)	Cohort study	Axelsson et al., 2010
E	Females, area with petrochemical industry vs. areas without petrochemical industry	1974–2005	1963	Age, sex	Sweden (Norum, located in central Stenungsund)		Exposure group: Female residents in area with petrochemical industry (N = N.R.) Reference group: Female residents without petrochemical industry (N = N.R.)			Cohort study	Axelsson et al., 2010
F	Residents aged < 65, petroleum	2000–2006	1958 (Oil refineries Ltd.)	Age, disease geographical incidence density	Israel (Kiryat Haim industrial zone)	Petroleum area: Areas with the centroid distance to the Kiryat Haim industrial zone	Exposure group: Residents aged < 65, living in petroleum area	Reported indicator: Relative density of cancer incident rate	Benzene, methylene chloride, acetaldehyde,	Cohort study	Zusman et al., 2012

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Table 1 (continued)

ID	Comparison	Study period	The started operation year of petrochemical plant	Adjusted confounders	Country	Definition of petroleum area and reference area	Study group (number of subject)	Outcome selection	Industrial activity/ substantial chemicals	Study design	Reference
	area vs. reference area		1974 (Gadiv Petrochemical Industries) 1985 (Gadot Chemical Tanks and Terminals Ltd.)			between 0 and 500 m Reference area: Areas with the centroid distance to the Kiryat Haim industrial zone 1000–1500 m	(N = 14,050) Reference group: Residents aged <65, residents in reference area (N = 22,780)	ICD version and code: N.R. Source: Israel National Cancer Registry	acrolein, and benzo[a]pyrene (BAP)		
G	Residents aged ≥ 65, petroleum area vs. reference area	2000–2006	1958 (Oil refineries Ltd.) 1974 (Gadiv Petrochemical Industries) 1985 (Gadot Chemical Tanks and Terminals Ltd.)	Age, disease geographical incidence density	Israel (Kiryat Haim industrial zone)		Exposure group: Residents aged ≥ 65, living in petroleum area (N = 2620) Reference group: Residents aged ≥ 65, residents in reference area (N = 3480)			Cohort study	Zusman et al., 2012
H	Males, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of Syracuse (excluding CS)	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Augusta, a municipality in the Province of Syracuse, Sicily)	Petroleum area: Four municipalities in the Province of Syracuse, Sicily, involving industrial activities, including refineries, petrochemical plants, liquid gas production, incinerator of harbor waste, cement plant, and landfills. Reference area: Province of Syracuse	Exposure group: Males living in Augusta (N = N.R.) Reference group: Males in the Province of Syracuse, excluding residents living in CSs (N = N.R.) (Exposure and reference group combined N = 396,517)	Reported indicator: Standardized incidence ratios ICD version and code: ICD-O-3, C33–34 Source: Cancer Registry of Syracuse Province	Heavy metals (arsenic, cadmium, chromium VI), polychlorinated biphenyls (PCBs), dioxins, PAHs, VOCs, BAP, nickel compounds	Cohort study	Fazzo et al., 2016
I	Females, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of Syracuse (excluding CS)	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Augusta, a municipality in the Province of Syracuse, Sicily)		Exposure group: Females living in Augusta (N = N.R.) Reference group: Females in the Province of Syracuse, excluding residents living in CSs (N = N.R.) (Exposure and reference group combined N = 396,517)			Cohort study	Fazzo et al., 2016
J	Males, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Melilli, a municipality in the Province of Syracuse, Sicily)		Exposure group: Males living in Melilli (N = N.R.) Reference group: Males in the Province of Syracuse, excluding			Cohort study	Fazzo et al., 2016

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Table 1 (continued)

ID	Comparison	Study period	The started operation year of petrochemical plant	Adjusted confounders	Country	Definition of petroleum area and reference area	Study group (number of subject)	Outcome selection	Industrial activity/substantial chemicals	Study design	Reference
	Syracuse (excluding CS)						residents living in CSs (N = N.R.)				
K	Females, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of Syracuse (excluding CS)	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Melilli, a municipality in the Province of Syracuse, Sicily)		(Exposure and reference group combined N = 396,517) Exposure group: Females living in Melilli (N = N.R.) Reference group: Females in the Province of Syracuse, excluding residents living in CSs (N = N.R.)			Cohort study	Fazzo et al., 2016
L	Males, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of Syracuse (excluding CS)	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Priolo, a municipality in the Province of Syracuse, Sicily)		(Exposure and reference group combined N = 396,517) Exposure group: Males living in Priolo (N = N.R.) Reference group: Males in the Province of Syracuse, excluding residents living in CSs (N = N.R.)			Cohort study	Fazzo et al., 2016
M	Females, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of Syracuse (excluding CS)	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Priolo, a municipality in the Province of Syracuse, Sicily)		(Exposure and reference group combined N = 396,517) Exposure group: Females living in Priolo (N = N.R.) Reference group: Females in the Province of Syracuse, excluding residents living in CSs (N = N.R.)			Cohort study	Fazzo et al., 2016
N	Males, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Syracuse, a municipality in the Province of Syracuse, Sicily)		(Exposure and reference group combined N = 396,517) Exposure group: Males living in Syracuse (N = N.R.) Reference group: Males in the Province of Syracuse, excluding			Cohort study	Fazzo et al., 2016

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Table 1 (continued)

ID	Comparison	Study period	The started operation year of petrochemical plant	Adjusted confounders	Country	Definition of petroleum area and reference area	Study group (number of subject)	Outcome selection	Industrial activity/substantial chemicals	Study design	Reference
	Syracuse (excluding CS)						residents living in CSs (N = N.R.)				
O	Females, residents in a municipality classified as a contaminated site (CS) vs. residents in the Province of Syracuse (excluding CS)	1999–2006	1949 (Augusta-Priolo petrochemical complex) (Italy)	Age, sex	Italy (Syracuse, a municipality in the Province of Syracuse, Sicily)		(Exposure and reference group combined N = 396,517) Exposure group: Females living in Syracuse (N = N.R.) Reference group: Females in the Province of Syracuse, excluding residents living in CSs (N = N.R.)			Cohort study	Fazzo et al., 2016
P	Males, residents living in the city with the largest oil refinery and petrochemical complex of the country	2003–2008	1959	Age, sex	Serbia (Pančevo)	Petroleum area: Pančevo (the southern part of the city has the largest oil refinery and petrochemical complex, named Southern Industrial Zone, of the country)	(Exposure and reference group combined N = 396,517) Exposure group: Males living in Pančevo (N = N.R.) Reference group: Males living in Central Serbia (N = N.R.)	Reported indicator: Age-adjusted incidence rates ICD version and code: ICD-10, C34	Benzene, toluene, methyl mercaptane, SO ₂ , NH ₃ , NO, CO, O ₃ , and PM ₁₀	Cohort study	Bulat et al., 2011
Q	Females, residents living in the city with the largest oil refinery and petrochemical complex of the country	2003–2008	1959	Age, sex	Serbia (Pančevo)	Reference area: Central Serbia	Exposure group: Females living in Pančevo (N = N.R.) Reference group: Females living in Central Serbia (N = N.R.)	Source: Cancer Registry of Vojvodina (for Pančevo) and Cancer Registry of the Serbian Institute of Public Health (for central Serbia)		Cohort study	Bulat et al., 2011

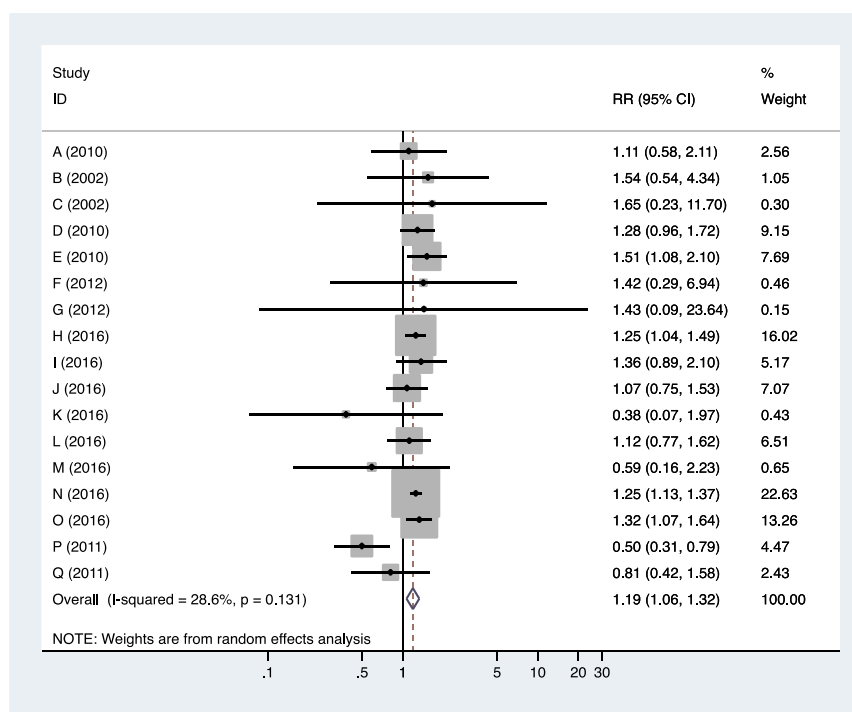


Fig. 2. Forest plot of studies on relative risks (RR) of lung cancer incidence among residents living nearby petrochemical industrial complexes.

not define exposure groups from the same community and failed to report the percentage of participants lost throughout the study.

Table 3 demonstrates estimated PAFs and number of lung cancer cases attributable to living in the vicinity of PICs in 22 EU countries. Previous review demonstrated approximately 9.6 million residents lived close to a PIC in 2010–2016. PAFs ranged widely among countries, from 0.003%–0.681% among males and 0.007%–1.630% among females. Spain and Germany had the highest number of estimated lung cancer cases attributable to PIC in 2016 among males (estimated 138 cases) and females (estimated 81 cases). A total of 494 male and 478 female lung cancer cases were attributable to PIC exposure in 22 EU countries every year.

Table 2

Subgroup analysis of relative risks of lung cancer incidence among residents living close to petrochemical industrial complexes, by different characteristics.

Characteristics	Numbers of studied groups	Pooled relative risks (95% CI)	I ² (%)	P-value
Overall	17	1.19 (1.06–1.32)	28.6	0.002
Sex				
Males	7	1.12 (0.95–1.33)	60.5	0.173
Females	7	1.29 (1.09–1.54)	5.1	0.004
Combined	3	1.16 (0.64–2.08)	0	0.627
Location				
United States	1	1.11 (0.58–2.11)	–	0.762
South Africa	2	1.56 (0.62–3.91)	0	0.342
Europe	12	1.17 (1.03–1.33)	50	0.019
Middle East	2	1.42 (0.36–5.66)	0	0.617
Bona fide observation				
≥ 7 years	12	1.25 (1.17–1.34)	0	< 0.001
< 7 years	5	0.85 (0.58–1.23)	21.0	0.381
Possible exposures				
VOC/PAH	10	1.24 (1.15–1.34)	0	< 0.001
Benzene	8	1.06 (0.74–1.53)	56.9	0.733
BAP	10	1.24 (1.15–1.33)	0	< 0.001
Heavy metals/PCB	8	1.24 (1.15–1.33)	0	< 0.001

4. Discussion

From 17 studied groups in six countries detailed in six published articles, we found residents living closer to PICs have 1.19-fold (95% CI = 1.06–1.32) higher risk of lung cancer compared to those living farther away from PICs. The 17 studied groups covered a large population (N = 466,066) with various characteristics, such as country, time period, and sex. Our further analysis of 9.64 million residents living close to PICs in 22 EU countries showed that a total of 972 lung cancer cases were attributable to residential exposure to PICs every year. Attributing up to 1.63% of lung cancers to merely living close to PICs

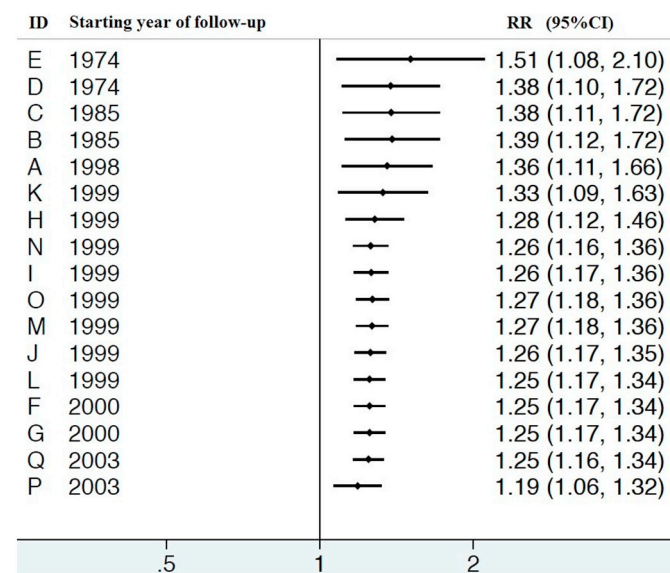


Fig. 3. Sensitivity analysis of pooled estimates by adding each additional study according to starting year of follow-up. RR = relative risk.

Table 3

Estimated population attributable factor and lung cancer incident cases attributable to living in the vicinity of petrochemical industry area in 22 European Union countries.

Country	Year	Exposed population	Population (million)	Males			Females		
				PAF	Lung cancer cases	Attributable cases	PAF	Lung cancer cases	Attributable cases
Austria	2016	17,409	8.63	0.02	3227.59 (3608.77–2869.88)	0.78 (0.87–0.69)	0.06	2055.71 (2320.38–1810.25)	1.2 (1.36–1.06)
Belgium	2015	513,600	11.27	0.54	6287.53 (7180.66–5539.67)	34.2 (39.06–30.13)	1.30	2511.96 (2800.56–2193.35)	32.76 (36.53–28.61)
Bulgaria	2016	412,684	7.22	0.68	2905.26 (3181.2–2623.41)	19.78 (21.66–17.86)	1.63	714.63 (785.04–649.47)	11.65 (12.79–10.58)
Croatia	2011	37,480	4.28	0.10	2448.27 (2742.31–2222.26)	2.57 (2.88–2.33)	0.25	899.46 (988.53–816.98)	2.28 (2.5–2.07)
Czech	2016	42,879	10.56	0.05	4418.88 (5013.82–3919.2)	2.15 (2.44–1.91)	0.12	2058.4 (2398.37–1790.46)	2.42 (2.82–2.11)
Denmark	2016	56,343	5.73	0.12	2444.24 (2718.13–2198.41)	2.88 (3.2–2.59)	0.28	2297.07 (2633.86–2029.73)	6.53 (7.49–5.77)
Finland	2015	69,124	5.5	0.15	1736.24 (1935.53–1588.04)	2.62 (2.92–2.39)	0.36	939.42 (1073.11–845.5)	3.41 (3.9–3.07)
France	2015	160,209	66.9	0.03	27,487.32 (30,491.96–25,066.72)	7.9 (8.76–7.2)	0.07	9752.16 (10,907.06–8698.81)	6.77 (7.57–6.04)
Germany	2015	1,258,199	82.67	0.16	40,287.78 (44,132.51–36,144.5)	63.9 (70–57.33)	0.38	21,108.84 (23,533.62–19,179.53)	80.73 (90–73.35)
Greece	2011	423,467	11.1	0.46	6123.43 (6830.34–5685.22)	27.91 (31.13–25.91)	1.09	1549.24 (1721.32–1413.81)	16.95 (18.84–15.47)
Hungary	2010	18,482	9.82	0.02	5979.92 (6340.93–5659.55)	1.35 (1.43–1.28)	0.05	3242.67 (3452.55–3039.32)	1.77 (1.88–1.66)
Ireland	2016	1153	4.77	0.00	1291.1 (1473.83–1132.47)	0.04 (0.04–0.03)	0.01	932.89 (1087.37–814.53)	0.07 (0.08–0.06)
Italy	2015	674,509	60.6	0.13	31,033.19 (34,957.96–27,875.42)	41.39 (46.63–37.18)	0.32	13,177.68 (14,601.22–11,659.34)	42.4 (46.98–37.51)
Lithuania	2014	35,634	2.87	0.15	1162.07 (1271.65–1076.72)	1.73 (1.89–1.6)	0.36	309.1 (342.45–277.69)	1.11 (1.23–1)
Netherlands	2016	634,068	16.69	0.45	9107.01 (10,346.41–7943.35)	41.33 (46.95–36.05)	1.09	5993.54 (6777.23–5276.34)	65.31 (73.85–57.5)
Poland	2015	580,900	37.95	0.18	16,244.49 (17,769.66–15,073.08)	29.78 (32.58–27.64)	0.44	6898.35 (7551.05–6388.13)	30.49 (33.37–28.23)
Portugal	2015	228,298	10.32	0.26	3031.48 (3219.9–2828.84)	8.03 (8.53–7.49)	0.64	934.12 (1020.85–858.48)	5.95 (6.51–5.47)
Romania	2012	908,800	19.71	0.41	8007.95 (8579.61–7441.97)	33.23 (35.6–30.88)	1.00	2039.5 (2208.81–1896.53)	20.33 (22.02–18.91)
Slovakia	2011	68,574	5.43	0.15	2115.49 (2366.75–1908.38)	3.2 (3.58–2.89)	0.36	665.99 (746–591.61)	2.43 (2.72–2.16)
Spain	2016	2,628,350	46.56	0.67	20,542.68 (23,378.57–18,218.53)	138.22 (157.3–122.58)	1.61	4689.54 (5273.8–4216.47)	75.53 (84.95–67.91)
Sweden	2016	490,000	9.9	0.60	2033.95 (2206.05–1879.27)	12.19 (13.22–11.26)	1.44	2087.6 (2265.41–1926.24)	29.98 (32.54–27.67)
UK	2016	385,177	65.64	0.07	26,997.96 (28,375.12–26,013.28)	19 (19.97–18.3)	0.17	22,074.46 (22,856.72–21,372.6)	37.5 (38.83–36.31)
Total		9,645,339	504.12		224,913.83 (248,121.67–204,908.17)	494.18 (550.64–445.52)		106,932.33 (117,345.31–97,745.17)	477.57 (528.76–432.51)

PAF: population attributable factor (unit: 100).

thus raises concerns of residential inequality, which warrants governments take advanced approaches to frame and strengthen air quality regulations for areas surrounding PICs.

We also discovered that females exposed to PICs have significantly greater risk of lung cancer, while males have a similar although non-significant trend. Previous studies identified that female smokers might be more susceptible to lung cancer (Risch et al., 1993; McDuffie et al., 1987), albeit occupational exposure might not play a role in gender differences (McDuffie et al., 1987). Our results generally indicate that residential exposure to PICs may increase residents' risk of lung cancer. Further, a sufficient study period (≥ 7 years) after a PIC has been in operation for 20 years renders significant estimation of lung cancer incidence among residents living near PICs. These results also are comparable to the previous assumption that latency period for lung cancer diagnosis usually requires approximately years to decades (Simonsen et al., 2010; Gottlieb et al., 1982). Our research further supported the greater risks of lung cancer incidence with residential exposure to specific chemicals related to PICs, such as VOC (Gao et al., 2014), PAH (Deziel et al., 2014), BAP (Gu et al., 2011), heavy metals (Park et al., 2012), and PCB (Carpenter, 2015).

Compared to our previous study analyzing the risk of lung cancer mortality in the vicinity of PICs for 13 groups in four countries (Lin et al., 2017), using incidence as the outcome can reduce bias caused by other competing risks, especially among the elderly. Also, incidence data is preferable to mortality data because it avoids other factors affecting lung cancer survival rates, such as healthcare quality (Checkoway et al., 2007). All incident cases in our studied groups were obtained from pathology samples and/or International Classification of

Diseases (ICD)-coded. Therefore, these results provide a better estimate of the risk epidemiology seeks to capture.

The fact that a total of 17 studied groups from six selected studies clustered in four regions might lead to concerns on generalizability. There are certain limitations in this study. First, some residual confounders might not be fully addressed in the studied groups, such as smoking. However, smoking is less prevalent among residents living in refinery counties (53.9%) than non-refinery counties (70.1%) (Fares and Masri, 2012). Since lower smoking rate is associated with lower lung cancer incidence (Hystad et al., 2013), we may have underestimated negative effects of PIC exposure given the smoking prevalence is similar in both groups. Other risk factors, such as socioeconomic status (Forrest et al., 2014) and education levels (Di Maio et al., 2012) might not be fully adjusted in the selected studies. Those confounders are more likely non-differential misclassification and bias the result toward the null. Further studies on susceptible groups or on synergistic risk factors are warranted. Second, some residents included in the study may have worked in nearby PICs and therefore been exposed to toxic chemicals in the workplace. Our study focuses on the incidence risk of lung cancer among residents living near PICs, which purposely includes both environmental exposure and occupational co-exposure. Since PIC workers unavoidably have a higher probability of living closer to PICs, estimating the overall risks for nearby residents provides strong and direct evidence that can be addressed by future policy. Third, possible underestimation of exposed populations in PICs might lead to an underestimation of PAFs and attributable cases. However, the EU Petroleum Refining Fitness Check has provided the best understanding of the exposed population by far (Lukach et al., 2015).

5. Conclusions

Analysis of current scientific literature suggests a significantly higher risk of lung cancer incidence is associated with living in the vicinity of PICs, particularly among female residents in Europe. Corresponding strategies are now urgently needed to reduce disease burden and address health concerns due to residential inequality.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and material

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

CKL contributed to study design, data analysis, reporting results, data interpretation, and drafting the manuscript. RTL contributed to study design, literature reviews, data acquisition, data interpretation, drafting the manuscript, and rewriting the manuscript. YTH contributed to literature reviews, data analysis, reporting results, data interpretation, and drafting the manuscript. DCC and HYH contributed to writing of the manuscript. All authors gave final approval of the version to be submitted and any revised version.

Declaration of interest

None.

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Appendix A. Supplementary data

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